

Tianfang Chang

✉ tfchang.ca@gmail.com | 📞 236-882-3992 | 🌐 Tianfang Chang | 📷 Tianfang Chang

Education

University of Victoria Sep. 2021 – Aug. 2024

Master of Applied Science - Electrical and Computer Engineering
CNLab (Computer Network Lab), Advisor: Lin Cai

Beijing University of Posts and Telecommunications Sep. 2015 – Jun. 2020

Bachelor of Engineering - Telecommunication Engineering

Skills

Programming Languages: C++, Python(Pytorch, Numpy, Matplotlib), Bash/Shell, HTML, SQL, REST API

Technologies & Tools: TCP/IP, NS-3, PyCharm, Docker, Git, Linux & Windows

Work Experience

Research Assistant - University of Victoria Jul. 2025 - Current

- Built a PyTorch Diffusion Transformer for satellite signal-strength prediction from geospatial data.

Research Assistant - University of Victoria Sep. 2021 - Aug. 2024

- Prototyped protocol components for next-generation networks in C++/Python using Network Simulator 3 (ns-3), improving efficiency and enabling new applications.

Teaching Assistant ECE363 - University of Victoria Jan. 2022 - Apr. 2023

- Designed and guided lab projects on network analysis. Mentored students on network project implementation.

Research Assistant - Institute of Computing Technology, CAS Aug. 2019 - Dec. 2019

- Optimized server resource scheduling using heuristic algorithms (ant colony, genetic, and simulated annealing) to maximize resource utilization.

Research Projects

Delay-guaranteed Routing Protocol  Github

Stack : C++, Python, ns-3, Linux, TCP/IP

- Designed and developed DDR, a novel distributed, traffic-aware routing protocol to provide packet-level delay guarantees for time-sensitive applications.
- Automated Linux/ns-3 experiments on ISP-grade graphs; traffic mixes covered mice/elephant/bursty flows; compared against ECMP/LFID/DGRP.
- Impact: reached near-100 % deadline hit rate under modest latency targets; under heavy/bursty load, DDR kept high on-time delivery while baselines degraded; initialization cost stayed lightweight (compute/memory only a small multiple of OSPF).

Multipath Routing Compatible TCP  Github

Stack : C++, Python, ns-3, Linux, TCP/IP

- Implemented an RTT-aware FastRTO to decouple reordering from loss, reducing false fast-retransmits on multipath.
- Built reproducible ns-3/Ubuntu experiments; benchmarked vs NewReno/CUBIC/BBR on OSPF (single-path) & ECMP (multipath) with real ISP topologies.
- Impact: on ECMP multipath, sustained 20–30 Mbps and 3–10× higher throughput than baselines; single-path parity with NewReno/CUBIC/BBR.

Publications

Multipath Routing Compatible Congestion Control Tianfang Chang , Lin Cai 2024 IEEE 100th Vehicular Technology Conference	VTC 2024
DDR: A Deadline-Driven Routing Protocol for Delay Guaranteed Service Pu Yang, Tianfang Chang , Lin Cai 2024 IEEE International Conference on Computer Communications	INFOCOM 2024
DGR: Delay-Guaranteed Routing Protocol Pu Yang, Lin Cai, Tianfang Chang 2023 IEEE 20th International Conference on Mobile Ad Hoc and Smart Systems	MASS 2023

Honors & Awards

Fellowship , University of Victoria Graduate Student	2024
Fellowship , University of Victoria Graduate Student	2023
Finalist , IEEE VTS Storytelling Competition	2023
Finalist , IEEE 2023 Workshop on Ubiquitous Poster Competition	2023
Fellowship , University of Victoria Graduate Student	2022
Scholarship , University of Victoria Graduate Student	2022
Fellowship , University of Victoria Graduate Student	2021

Community Activity

Paper Reviewer IEEE INFOCOM 2023 - 2024, IEEE Network Magazine 2023 - 2024	2022-2023
Volunteer Volunteer of IEEE 2023 Workshop on Ubiquitous	2023
Volunteer Volunteer of Explore UVIC Event	2023